



Sydney Girls High School

2002

TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION

Mathematics

General Instructions

- Reading Time - 5 mins
- Working time - 3 hours
- Attempt ALL questions
- ALL questions are of equal value
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Standard integrals are supplied
- Board-approved calculators may be used.
- Diagrams are not to scale
- Each question attempted should be started on a new sheet. Write on one side of the paper only.

This is a trial paper ONLY. It does not necessarily reflect the format or the contents of the 2002 HSC Examination Paper in this subject.

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Candidate Number

Question 1.

- (a) Solve for x
 $(2x - 5)(x + 2) = 0$
- (b) If $x = 4$, evaluate $\frac{e^x - 1}{e^x + 1}$ correct to 3 significant figures
- (c) Solve for x
 $|x + 5| = 3$
- (d) Simplify $\frac{3x}{5} - \frac{x-1}{2}$
- (e) Express $0.\dot{3}\dot{1}$ as a fraction in lowest terms.

Question 2.

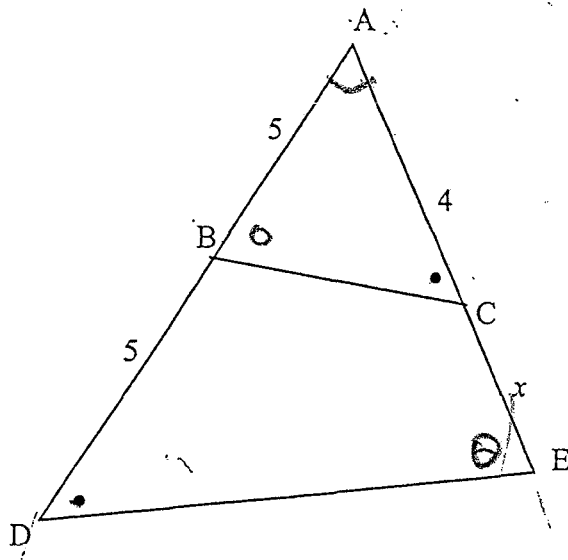
A (2,3) B (-2, -7)

- (a) Find
- (i) length of AB ✓
 - (ii) midpt of AB ✓
 - (iii) gradient of AB ✓
 - (iv) equation of AB ✓
 - (v) y - intercept of AB ✓
- (b) (i) Find the equation of the line perpendicular to AB passing through (2,5)
- (ii) What angle does this line make with the x-axis?
- (c) Graph the region on the number plane for which $y > 3$ and $x + y \leq 3$

0.23

Question 3.

(a)

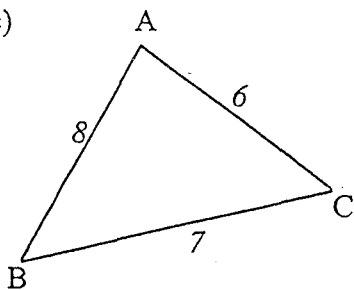


check

Find x

- (b) Write the exact value of
- (i) $\cos 30^\circ$
 - (ii) $\tan 330^\circ$

(c)



Use the cos rule to find $\angle B$
correct to the nearest minute.

Question 4.

- (a) Solve for x :

$$(x-2)(x+5) = 8$$

- (b) Use the quadratic formula to find x correct to 1 decimal place:

$$2x^2 - x - 14 = 0$$

- (c) (i) Graph on the number plane the piecemeal function

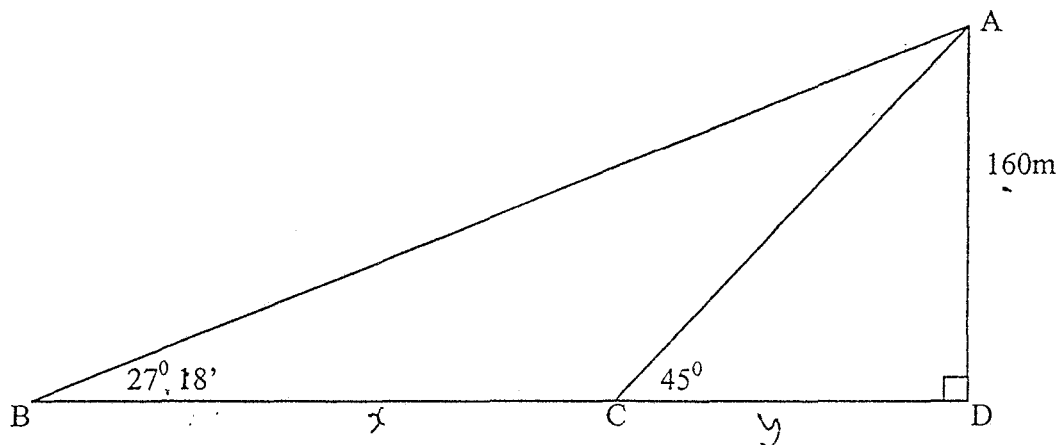
$$f(x) = \begin{cases} |2x| & \text{for } x \leq -2 \\ 4 & \text{for } -2 < x < 2 \\ 2x & \text{for } x \geq 2 \end{cases}$$

- (ii) State whether the function $y = f(x)$ is odd, even or neither

- (iii) State whether the function $y = f(x)$ is continuous or discontinuous.

Question 5.

- (a)



An observer in a boat rows towards a cliff which is 160m high.

At point B, the angle of elevation of the cliff top is $27^\circ 18'$

At point C, the angle of elevation of the cliff top is 45°

What is the distance BC to the nearest metre?

- (b) The area enclosed by the curve $y = x^2$, the y -axis and the line $y = a$ is exactly $10 u^2$. Find a to 2 decimal places.

- (c) What is the primitive of e^x ?

Question 6.

(a) Find $\frac{dy}{dx}$ for

(i) $y = \frac{x-2}{x+4}$

(ii) $y = e^{2x}$

(iii) $y = \sin 5x$

(iv) $y = \log_e (\tan x)$

(b) Find the equation of the tangent to the curve $y = x^2 - x$ at the point (2, 2)

(c) Locate the stationary points on $y = x^3 - 3x$ and determine their nature.

Question 7.

(a)

Seeds from a particular plant have 2 chances in 5 of surviving to flowering stage. $\frac{3}{4}$ of the surviving plants bear pink flowers and $\frac{1}{4}$ bear white flowers. Three seeds are planted. What is the probability:

- (i) none survive
(ii) all 3 survive and bear pink flowers
(iii) at least one pink-flowering plant survives

ch

(b) \$1000 is deposited in an account that pays 5%pa interest.

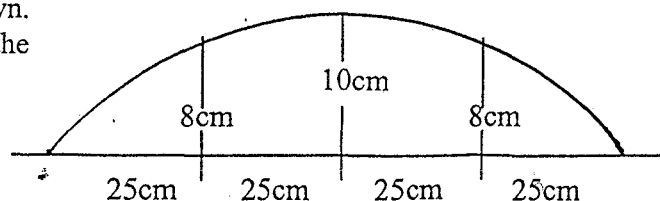
- (i) How much will it be worth after 15 years?
(ii) If \$1000 is deposited at the beginning of each year into an account that pays 5% pa interest, what will be the balance at the end of 15 years.

(c)

A speed bump has cross-section as shown. Use Simpson's Rule to find the area of the cross-section.

Hence find the volume of concrete required for a speed-bump for a road 10m wide.

Answer in cubic metres.



x10

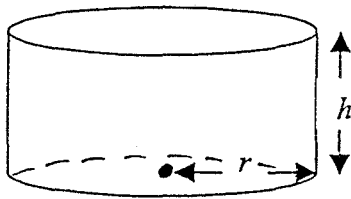
Question 8.

(a) Find the value(s) of k for which $x^2 - (k+2)x + (4k-4) = 0$

- has
- (i) two equal roots
 - (ii) one root the reciprocal of the other
 - (iii) one root equal to 8

(b) Solve for x : $4^x - 7 \cdot 2^x - 8 = 0$

(c)



Given $V = \pi r^2 h$, and $r + h = 12 \text{ cm}$

Show that $\frac{dv}{dr} = 3\pi r(8-r)$

and hence show that the maximum volume of the cylinder is $256\pi \text{ cm}^3$

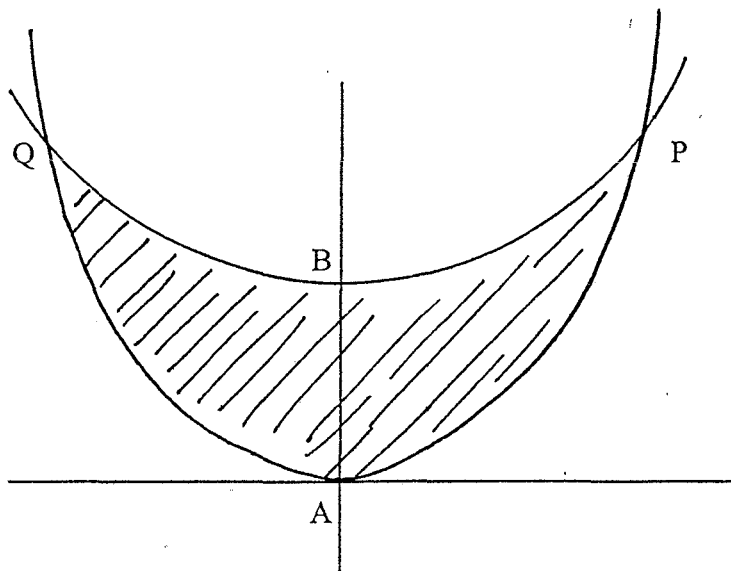
Question 9.

- (a) A curve of $y = f(x)$ passes through the point $(2, 5)$ and $(3, a)$.
If its gradient function is given by $f'(x) = 3x^2 - 4x$
Find the value of a .

- (b) A parabola has focus at $(2, 4)$ and vertex at $(2, -2)$
Find
- (i) its focal length
 - (ii) the equation of the parabola
 - (iii) the equation of its axis of symmetry
 - (iv) the equation of its directrix

- (c)
- (i) By completing the square, or otherwise, find the centre of the circle with equation $x^2 + 6x + y^2 - 2y = 15$
 - (ii) Find algebraically the points A and B where the circle cuts the y-axis
 - (iii) What is the area of $\triangle ABC$?

Question 10.



- (a) The two curves $y = 2x^2$ and $y = x^2 + 1$ intersect at P and Q as shown.
- Find the coordinates of B, P and Q.
 - The shaded region is rotated about the y-axis to form a solid bowl shape. Find the volume of the bowl in terms of π
- (b) Sketch the curve $y = 2\sin x$ $0 \leq x \leq 2\pi$
Find the area enclosed by the curve and the x-axis
- (c) The curve $y = 1 + \tan x$ between $x = 0$ and $x = \frac{\pi}{4}$ is rotated about the x-axis. Find the volume of the solid so formed.

-- END OF PAPER --

SYDNEY GIRLS HIGH SCHOOL - 2002 MATHEMATICS
TRIAL

Question 1

a) $2x - 5 = 0 \quad x + 2 = 0$
 $2x = 5 \quad x = -2$
 $x = \frac{5}{2}$
 $\therefore x = \frac{5}{2}, -2$

b) $\frac{e^4 - 1}{e^4 + 1} = \frac{53.59815}{55.59815} = 0.964 \text{ (3 sig figs)}$

c) $x + 5 = 3 \quad x + 5 = -3$
 $x = -2, x = -8$

d) $\frac{6x - 5(x-1)}{10}$
 $= \frac{6x - 5x + 5}{10}$
 $= \frac{x + 5}{10}$

12

e) let $x = 0.\dot{3}\dot{1}$
 $10x = 23.\dot{1}\dot{3}$
 $100x = 31.\dot{3}\dot{1}$
 $\therefore 99x = 31$
 $x = \frac{31}{99}$

Question 2

a) (i) $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad A(2, 3) \quad B(-2, -7)$
 $= \sqrt{(-2 - 2)^2 + (-7 - 3)^2}$
 $= \sqrt{116} = \sqrt{4 \times 29}$
 $= 2\sqrt{29}$

(ii) $\left(\frac{2 + (-2)}{2}, \frac{3 + (-7)}{2} \right) = (0, -2)$

(ii) $A(2,3) B(-2,-7)$

$$m_{AB} = \frac{-7-3}{-2-2}$$

$$= \frac{-10}{-4} = \frac{10}{4} = \frac{5}{2} \checkmark$$

(iv) $y-3 = \frac{5}{2}(x-2)$

$$y-3 = \frac{5x-10}{2} \checkmark$$

$$2y-6 = 5x-10$$

$$5x-2y-10+6=0$$

$$5x-2y-4=0 \checkmark$$

12

(v) y intercept, when $x=0$

$$0-2y-4=0$$

$$-2y=4$$

$$y=-2$$

$$\therefore y \text{ intercept is } (0,-2) \checkmark$$

b) (i) perpendicular to AB i.e.

$$m_{AB} \times m_{\perp AB} = -1$$

$$\therefore \frac{5}{2} \times m = -1$$

$$m = -1 \div \frac{5}{2}$$

$$= -\frac{2}{5} \checkmark$$

$$\therefore y-5 = -\frac{2}{5}(x-2)$$

$$y-5 = -\frac{2x+4}{5}$$

$$5y-25 = -2x+4 \checkmark$$

$$2x+5y-29=0$$

(ii) $5y = -2x + 29$

$$y = -\frac{2x}{5} + \frac{29}{5}$$

$$\therefore m = -\frac{2}{5}$$

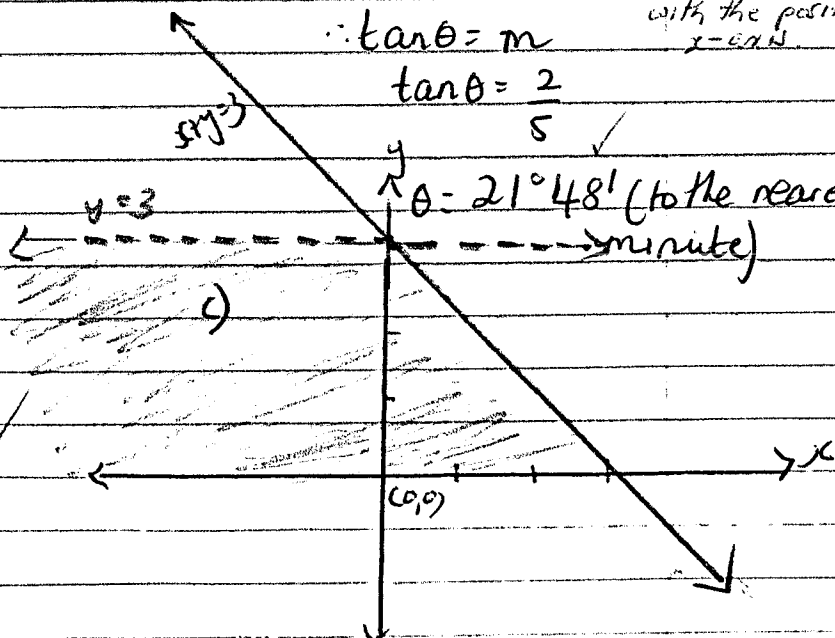
$$\theta = 158^\circ 12'$$

with the positive x -axis.

$$\therefore \tan \theta = m$$

$$\tan \theta = -\frac{2}{5} \checkmark$$

$$\theta = 21^\circ 48' \text{ (to the nearest minute)}$$



Question 3

a) In Δ 's ABC and ADE

$\angle A$ is common

$\angle ACB = \angle ADE$ (given)

$\therefore \Delta ABC \sim \Delta ADE$ (equiangular)

$\therefore \frac{AD}{AB} = \frac{AE}{AC}$ (corresponding sides in similar Δ 's)

$\frac{10}{5} = \frac{x+4}{4}$

$$\therefore \frac{10}{5} = \frac{x+4}{4}$$

$$\frac{10}{5} = \frac{x+4}{4}$$

$$40 = 5x + 20$$

$$50 = 16 + 4x$$

$$5x = 20$$

$$34 = 4x$$

$$x = 4 \text{ units}$$

$$\frac{34}{4} = x \Rightarrow x = 8.5$$

10

b) (i) $\cos 30^\circ = \frac{\sqrt{3}}{2}$ ✓

(ii) $\tan 30^\circ = \frac{1}{\sqrt{3}}$ ✓

c) $\cos B = \frac{8^2 + 7^2 - 6^2}{2 \times 8 \times 7}$

$$\cos B = \frac{77}{112}$$

$$\cos B = 0.6875$$

$$\therefore \angle B = 46^\circ 34' \text{ (to the nearest minute)}$$

Question 4

a) $x^2 + 5x - 2x - 10 = 8$

$$x^2 + 3x - 10 = 8$$

$$x^2 + 3x - 10 - 8 = 0$$

$$x^2 + 3x - 18 = 0$$

$$(x-3)(x+6) = 0$$

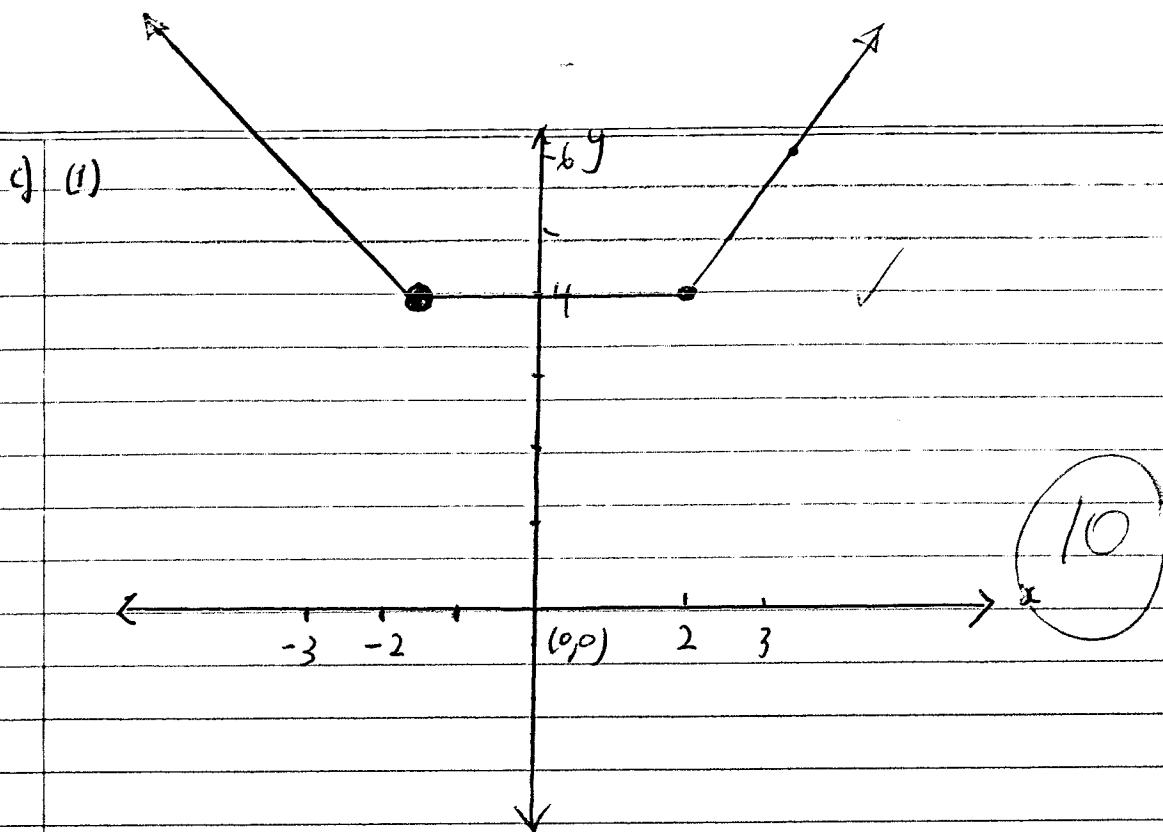
$$x=3, x=-6$$

b) $2x^2 - x - 14 = 0$

$$x = \frac{1 \pm \sqrt{1 - 4 \times 2 \times -14}}{4}$$

$$x = \frac{1 \pm \sqrt{113}}{4}$$

$$\therefore x = 2.9, x = -2.4 \text{ (to 1 d.p.)}$$



(i) even function ✓

(ii) ~~discontinuous~~ continuous

Question 5

a) $\angle ACB = 135^\circ$ (angle sum of straight line is 180°)
 $\therefore \angle BAC = 17^\circ 42'$ (angle sum of triangle = 180°)
 $\angle CAD = 45^\circ$ (angle sum of $\Delta = 180^\circ$)
 $\therefore \Delta ADC$ is isosceles

$$\therefore CD = 160\text{m}$$

$$\sin 45^\circ = \frac{160}{AC}$$

$$AC = \frac{160}{\frac{1}{\sqrt{2}}}$$

$$160 \div \frac{1}{\sqrt{2}}$$

$$160 \times \sqrt{2}$$

$$AC = 160\sqrt{2} \text{ metres}$$

In ΔABC

$$\frac{BC}{\sin 17^\circ 42'} = \frac{AC}{\sin 27^\circ 18'} \quad (\text{Sine rule})$$

Given to find BD

$$\text{using } \tan 27^\circ 18' = \frac{160}{BC + 160}$$

$$\therefore BC + 160 = \frac{160}{\tan 27^\circ 18'}$$

$$\rightarrow BC = \frac{AC \sin 17^\circ 42'}{\sin 27^\circ 18'} \Rightarrow BC = 149.99 = 150\text{m}$$

$$AC = 160\sqrt{2}$$

$$\therefore BC = \frac{160\sqrt{2} \times \sin 17^\circ 42'}{\sin 27^\circ 18'}$$

$$\therefore BC = 149.99 \dots \dots = 150\text{m (to the nearest metre)}$$

b) $y = x^2$
 $x^2 = y$
 $x = \pm \sqrt{y}$

$$\therefore \int_0^a y^{1/2} dy = \left[\frac{y^{3/2}}{3/2} \right]_0^a = 10$$

$$\sqrt{y^3} \div \frac{3}{2}$$

$$\sqrt{y^3} \times \frac{2}{3}$$

$$\left[\frac{\pm 2\sqrt{y^3}}{3} \right]_0^a = 10$$

12

$$\pm \frac{2\sqrt{a^3}}{3} - \frac{2\sqrt{0}}{3} = 10$$

$$\pm \frac{2\sqrt{a^3}}{3} = 10 \quad \checkmark$$

$$\therefore \frac{2\sqrt{a^3}}{3} = 10$$

$$\frac{-2\sqrt{a^3}}{3} = 10$$

$$2\sqrt{a^3} = 30$$

$$-2\sqrt{a^3} = 30$$

$$\sqrt{a^3} = 15$$

$$\sqrt{a^3} = -15$$

$$a^3 = 225$$

$$a^3 = 225$$

$$a = 6.08 \text{ (to 2 d.p.)}$$

↑ same solution

$$\therefore a = 6.08$$

c) $\int e^x dx = e^x + c \quad \checkmark$

Question 6

a) (1) $u = x-2 \quad v = x+4$

$$u' = 1 \quad v' = 1$$

$$\therefore \frac{dy}{dx} = \frac{x+4 - (x-2)}{(x+4)^2} = \frac{x+4 - x+2}{(x+4)^2} = \frac{6}{(x+4)^2} \quad \checkmark$$

$$(i) \quad y = e^{2x}$$

$$\frac{dy}{dx} = 2e^{2x} \quad \checkmark$$

$$(ii) \quad y = \sin 5x$$

$$\frac{dy}{dx} = \cos 5x \times 5$$

$$\frac{dy}{dx} = 5 \cos 5x \quad \checkmark$$

$$(iv) \quad \frac{dy}{dx} = \frac{\sec^2 x}{\tan x} \quad \checkmark$$

$$b) \quad y = x^2 - x$$

$$\frac{dy}{dx} = 2x - 1 \quad \checkmark$$

$$\text{at } x=2, m_T = 2 \times 2 - 1$$

$$= 3 \quad \checkmark$$

$$\therefore y - 2 = 3(x - 2)$$

$$y - 2 = 3x - 6 \quad \checkmark$$

$$3x - y - 4 = 0$$

12

$$c) \quad y = x^3 - 3x$$

$$\frac{dy}{dx} = 3x^2 - 3$$

to find stationary points, let $\frac{dy}{dx} = 0$

$$\therefore 3x^2 - 3 = 0$$

$$\cancel{3} + 3(x^2 - 1) = 0$$

$$3(x+1)(x-1) = 0 \quad \checkmark$$

$$\therefore x = \pm 1$$

$$\text{when } x=1, y = -2$$

$$\text{when } x=-1, y = 2 \quad \checkmark$$

$\therefore$ stationary points are $(1, -2)$ and $(-1, 2)$

$$y'' = 6x$$

at $x=1, y'' = 6 > 0 \therefore (1, -2)$ is a minimum turning point

at $x=-1, y'' = -6 < 0 \therefore (-1, 2)$ is a maximum turning point

$$b) (i) A = P \left(1 + \frac{r}{100}\right)^n$$

$$A = 1000 \left(1 + \frac{5}{100}\right)^{15}$$

$$A = 1000(1.05)^{15} \checkmark$$

$$= \$2078.93 \text{ (to 2 dp)}$$

9

$$(ii) A_1 = 1000(1.05)^1$$

$$+ 1000(1.05)^2$$

$$A_2 = 1000(1.05) + 1000(1.05)^2$$

$$= 1000(1.05 + 1.05^2)$$

$$\therefore A_{15} = 1000(1.05 + 1.05^2 + 1.05^3 + \dots + 1.05^{15}) \checkmark$$

G.P

$$\text{with } a = 1.05, r = 1.05, n = 15$$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_n = \frac{1.05(1.05^{15} - 1)}{0.05}$$

$$\therefore A_{15} = 1000 \left(\frac{1.05(1.05^{15} - 1)}{0.05} \right) \checkmark$$

$$\$22657.49 \text{ (to 2 dp)} \checkmark$$

c)

x	0	25	50	75	100
y	0	8	10	8	0
	1st	4	2	4	Last

$$h = 25$$

$$A \doteq \frac{25}{3} [0 + 0 + 4(8 + 8) + 2(10)] \checkmark$$

$$\text{Area} \doteq 700 \text{ cm}^2$$

$$\therefore \text{Volume} = 700 \times 10$$

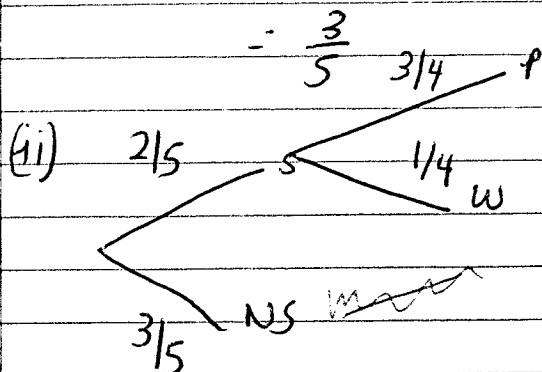
$$= \underline{7000 \text{ cm}^3} \checkmark$$

Question 7

a) (i) $P(\text{surviving}) = \frac{2}{5}$

$$P(\text{not surviving}) = 1 - \frac{2}{5}$$

$$= \frac{3}{5}$$



$$P(SP) = \frac{2}{5} \times \frac{3}{4} = \frac{3}{10}$$

three ~~flower~~ plants = $\frac{3}{10} \times \frac{3}{10} \times \frac{3}{10}$

$$= \frac{9}{1000}$$

$$= \frac{27}{1000}$$

(iii) $\frac{1 - P(\text{non-surviving})}{1 - 1}$

$$1 - P(\text{None surviving})$$

$$= 1 - \frac{27}{125}$$

$$= \frac{98}{125}$$

Question 8

a) (i) $\Delta = (k+2)^2 - 4 \times 1 \times (4k-4)$

$$k^2 + 4k + 4 - 4(4k-4)$$

$$k^2 + 4k + 4 - 16k + 16$$

$$\Delta = k^2 - 12k + 20$$

equal roots, i.e. $\Delta = 0$

$$\therefore k^2 - 12k + 20 = 0$$

$$(k-2)(k-10) = 0$$

$$\therefore k=2 \quad k=10$$

(ii) let roots be α and $\frac{1}{\alpha}$

$$\alpha \times \frac{1}{\alpha} = 1$$

$$\therefore \frac{4k-4}{1} = 1$$

$$4k-4=1$$

$$4k=5$$

$$k = \frac{5}{4}$$

(iii) Substg $x=8$

$$64 - (k+2) \times 8 + (4k-4) = 0$$

$$64 - 8(k+2) + 4k - 4 = 0$$

$$64 - 8k - 16 + 4k - 4 = 0$$

$$44 - 4k = 0$$

$$4k = 44$$

$$k = 11$$

b) let $u = 2^x$

$$\therefore u^2 - 7u - 8 = 0$$

$$(u+1)(u-8) = 0$$

$$\therefore u = -1, \quad u = 8$$

$$\therefore 2^x = -1, \quad 2^x = 8$$

$$\text{no solutions} \quad 2^x = 2^3$$

$$\therefore \text{no solutions} \quad \therefore x = 3$$

$\therefore$ the only solution is $x = 3$

$\rightarrow$ c) $h = 12 - r$

$$V = \pi r^2 h$$

$$= \pi r^2 (12 - r)$$

$$12\pi r^2 - \pi r^3$$

$$\therefore \frac{dV}{dr} = 24\pi r - 3\pi r^2$$

to find stationary points

$$\text{let } \frac{dV}{dr} = 0$$

$$\therefore 3\pi r(8-r) = 0$$

$$3\pi r = 0, \quad 8-r = 0$$

$$r = 0, \quad r = 8$$

$r \neq 0$ as it is a length

$$\therefore r = 8 \text{ cm}$$

$$u = 2\pi r, \quad v = 8 - r$$

$$u' = 2\pi, \quad v' = -1$$

$$\frac{d^2V}{dr^2} = 24\pi - 6\pi r$$

$$\text{at } r = 8, \quad \frac{d^2V}{dr^2} = 24\pi - 48\pi = -24\pi$$

$$\frac{dV}{dr} = 24\pi r - 3\pi r^2$$

$$\frac{d^2V}{dr^2} = 24\pi - 6\pi r$$

$$\text{at } r = 8, \quad \frac{d^2V}{dr^2} = -24\pi < 0$$

a maximum.

$$V = \pi r^2 h, \quad h + r = 12, \quad h = 4$$

12

$$\therefore V = \pi r^2 h$$

$$= \pi \times 8 \times 8 \times 4$$

$$= 256\pi \text{ cm}^3 \quad \checkmark$$

Question 9

$$\int 3x^2 - 4x \, dx = \frac{3x^3}{3} - \frac{4x^2}{2} + c$$

$$f(x) = x^3 - 2x^2 + c$$

passes through (2, 5) $\checkmark$

$$\therefore 5 = c + 0$$

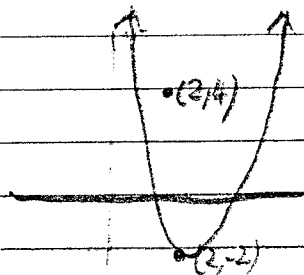
$$\therefore c = 5$$

$$\therefore f(x) = x^3 - 2x^2 + 5 \quad \checkmark$$

passes through (3, a)

$$\therefore a = 14 \quad \checkmark$$

b) (1)



focal length = 6 units $\checkmark$

(i) ~~14~~ ~~14~~

$$(x-2)^2 = 24(y+2) \quad \checkmark$$

$$4a = 24$$

$$a =$$

$$(ii) \quad x = 2 \quad \checkmark$$

$$(iv) \quad y = -8 \quad \checkmark$$

$$c) \quad (i) \quad x^2 + 6x + 9 + y^2 - 2y + 1 = 15 + 1 + 9$$

$$(x+3)^2 + (y-1)^2 = 25$$

$$\therefore \text{centre} = (-3, 1) \quad \checkmark$$

(ii) cuts y axis when $x = 0$

$$\therefore (0+3)^2 + (y^2 - 2y + 1) = 25$$

* Try this

$$3^2 + (y-1)^2 = 25$$

$$(y-1)^2 = 16 \Rightarrow y-1 = \pm 4$$

$$y = 5 \text{ or } -3$$

$$9 + y^2 - 2y + 1 = 25$$

$$10 + y^2 - 2y = 25$$

*
Te = 10y

$$y^2 - 2y + 10 - 25 = 0$$

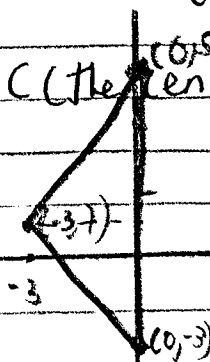
$$y^2 - 2y - 15 = 0$$

$$(y+3)(y-5) = 0$$

$$y = -3, y = 5$$

$\therefore$ A and B are (0, -3) $\checkmark$
and (0, 5)

(ii) C (the centre)



AB = 8 units

height = 3 ✓

$$\frac{1}{2} \times 3 \times 8$$
$$= \frac{24}{2}$$

$$= 12 \text{ units}^2$$

12

Question 10

a) (i) intersection: $2x^2 = x^2 + 1$

$$x^2 = 1$$

$$x^2 - 1 = 0$$

$$(x+1)(x-1) = 0$$

$$\therefore x = \pm 1$$

when $x = 1, y = 2$

when $x = -1, y = 2$

$\therefore Q(-1, 2)$ and $P(1, 2)$ ✓

$$y = x^2 + 1$$

at B, $x = 0$

$$\therefore y = 1$$

$\therefore B(0, 1)$ ✓

$\therefore B(0, 1); P(1, 2)$ and $Q(-1, 2)$

(ii) $y = 2x^2$

$$2x^2 = y$$

$$x^2 = \frac{y}{2}$$

$$\frac{y}{2} - (y-1)$$

$$y = x^2 + 1$$

$$y - 1 = x^2$$

$$x^2 = y - 1$$

$$\therefore V = \pi \int_0^1 \frac{y}{2} - (y-1) dy$$

$$\frac{y}{2} - y + 1$$
$$\frac{y - 2(y+1)}{2}$$

$$\frac{y - 2y - 2}{2}$$

$$V = \pi \int_0^1 \frac{y - 2y + 2}{2} dy$$

$$\frac{\pi}{2} \int_0^1 y - 2y + 2 dy = \frac{y^2}{2} - \frac{2y^2}{2} + 2y$$

$$\frac{\pi}{2} \left[\frac{y^2}{2} - y^2 + 2y \right]_0^1$$

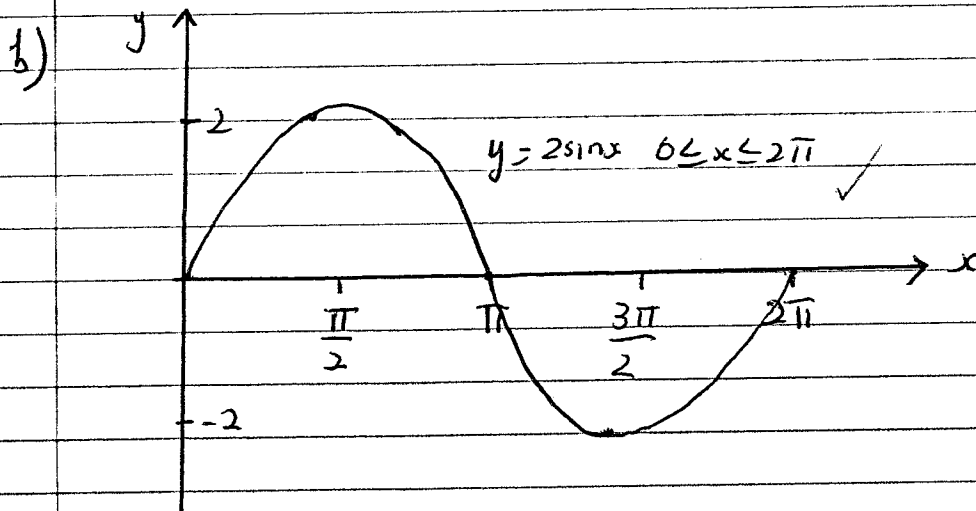
$$\left[\left(\frac{1}{2} - 1 + 2 \right) \right] - 0$$

$$= \frac{5}{2} \quad \frac{3}{2} \times \frac{\pi}{2}$$

$$\frac{5}{2} \times \frac{\pi}{2} = \frac{5\pi}{4} \text{ units}^3$$

$$\text{Total volume} = 2 \times \frac{5\pi}{4}$$

$$= \frac{10\pi}{4} = \frac{5\pi}{2} \text{ units}^3$$



$$\text{Area} = 2 \times \int_0^{\pi} 2 \sin x \, dx = \left[-2 \cos x \right]_0^{\pi} \times 2$$

$$[2 - -2] = 4$$

$$4 \times 2 = 8 \text{ units}^2$$

$$c) \frac{\pi}{4} \int_0^{\pi/4} (1 + \tan x)^2 \, dx = (1 + \tan x)(1 + \tan x)$$

$$= \pi \int_0^{\pi/4} 1 + 2 \tan x + \tan^2 x \, dx \quad \checkmark$$

$$= \pi \int_0^{\pi/4} 1 + 2 \tan x + \sec^2 x - 1 \, dx \quad \checkmark$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$\tan^2 \theta = \sec^2 \theta - 1$$

$$\pi \int_0^{\pi/4} 1 + 2 \left(\frac{\sin x}{\cos x} \right) + \sec^2 x - 1 \, dx = \pi \left[x + -2 \ln(\cos x) + \tan x - x \right]_0^{\pi/4}$$

$$\pi \left[\cancel{x} - 2 \ln(\cos x) + \tan x - \cancel{x} \right]_0^{\pi/4} \checkmark$$

$$\pi \left[-2 \ln(\cos x) + \tan x \right]_0^{\pi/4}$$

$$\pi \left[-2 \ln\left(\frac{1}{\sqrt{2}}\right) + 1 \right] - [0] \checkmark$$

$$= 2\pi \left(\ln\left(\frac{1}{\sqrt{2}}\right) + 1 \right) \text{ units}^3$$

$$\pi \left[-2 (\ln 1 - \ln \sqrt{2}) + 1 \right]$$

$$= \pi \left[2 \ln \sqrt{2} + 1 \right]$$

$$= \pi (\ln 2 + 1) \text{ units}^3$$

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